

ORIGINAL RESEARCH

Empagliflozin Improves Angina and Endothelial Dysfunction Indices in Angina and No Obstructive Coronary Arteries (ANOCA)

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BACKGROUND: Treatments targeting endothelial dysfunction in patients with angina and no obstructive coronary arteries (ANOCA) and coronary microvascular and vasomotor dysfunction (CMVD) are lacking. We evaluated the effects of SGLT2i (sodium-glucose cotransporter-2 inhibition) on angina, inflammatory and oxidative stress biomarkers, and endothelial dysfunction indices using a serum-induced in vitro model in ANOCA patients with CMVD. We conducted an open-label pilot trial of 11 ANOCA participants with CMVD diagnosed by invasive coronary function testing receiving empagliflozin (10 mg daily) for 90 days.

METHODS: Angina (Seattle Angina Questionnaire), circulating inflammatory and oxidative stress biomarkers, and endothelial microvesicles (eEVs) were assessed from baseline to 45 and 90 days, and from 90 days to posttreatment discontinuation in a subset (N=5). A serum-induced in vitro model of human coronary artery endothelial cells was used to evaluate endothelial cell apoptosis, viability, eEV release, VCAM-1 (vascular cell adhesion molecule-1) expression, and mitochondrial reactive oxygen species. Stored serum from ANOCA patients without CMVD (controls, N=6) was compared with baseline samples.

RESULTS: All trial participants had coronary endothelial dysfunction, with concomitant coronary microvascular dysfunction (55%), coronary spasm (54%), and microvascular spasm (18%). Empagliflozin significantly improved Seattle Angina Questionnaire summary scores by 14.0 (4.2–23.9) points at 90 days ($P<0.001$). Compared with baseline, empagliflozin significantly reduced inflammatory cytokines, oxidative stress biomarkers, and inflammatory eEVs while increasing anti-inflammatory cytokines, antioxidants, and repair eEVs at 45 and 90 days; these effects rebounded after treatment discontinuation. Similarly, serum collected during empagliflozin treatment significantly reduced endothelial cell apoptosis, eEV release, VCAM-1 expression, and mitochondrial reactive oxygen species in vitro, with reversal of these effects at posttreatment discontinuation.

CONCLUSIONS: In ANOCA participants with invasive coronary function testing confirmed CMVD, SGLT2i therapy produced clinically meaningful anti-anginal effects and reversed multiple indices of endothelial dysfunction, with rebound after treatment discontinuation. These findings support further evaluation in larger clinical trials.

Key Words: biomarkers ■ blood pressure ■ cytokine ■ heart failure ■ oxidative stress

Between 50 and 70% of all individuals presenting for invasive coronary angiography each year, estimated at 1 million cases per year in the United States, have angina/ischemia with nonobstructive coronary arteries (ANOCA).^{1–3} In the majority of ANOCA patients with

ischemia with nonobstructive coronary arteries, angina and ischemia are due to underlying coronary microvascular and vasomotor dysfunction (CMVD) associated with a 4-fold increased risk of all-cause mortality.^{4–7} Although the pathophysiologic mechanisms of CMVD are not

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Nonstandard Abbreviations and Acronyms

AE	adverse event
ANOCA	angina with no obstructive coronary arteries
CAD	coronary artery disease
CED	coronary endothelial dysfunction
CMVD	coronary microvascular and vasomotor dysfunction
CRP	C-reactive protein
DASI	Duke Activity Status Index
EC	endothelial cell
eEV	endothelial microvesicle
EnD	endothelial dysfunction
EV	extracellular vesicle
HCAEC	human coronary artery endothelial cell
ICAM-1	intercellular adhesion molecule-1
ICFT	invasive coronary functional testing
IL	interleukin
METS	metabolic equivalents
MPO	myeloperoxidase
PTD	post treatment discontinuation
ROS	reactive oxygen species
SAQ	Seattle Angina Questionnaire
SGLT2i	sodium glucose cotransporter-2 inhibitor
T2DM	type 2 diabetes
TNFα	tumor necrosis factor- α
VCAM-1	vascular cell adhesion molecule-1
VEGF	vascular endothelial growth factor

well known, inflammation, oxidative stress, and endothelial dysfunction (EnD) are hypothesized to play pivotal roles.^{8–12} Current treatments for patients with ANOCA are predominantly limited to antianginal medications that do not target the underlying CMVD.

SGLT2i (sodium glucose cotransporter 2 inhibitors) are a novel class of drugs initially developed as potent glycemic control medications for patients with type 2 diabetes (T2DM).¹³ Several large, randomized, double-blind, placebo-controlled clinical trials (EMPA-REG and EMPEROR-Preserved) have demonstrated striking cardiovascular benefit of SGLT2i in populations with a high prevalence of CMVD, including patients with T2DM and heart failure with preserved ejection fraction, although the mechanisms underlying these benefits remain unclear.^{14–16} Although SGLT2i effects on patients with ANOCA with CMVD have not been studied to date, animal studies show improvements in CMVD endotypes with SGLT2i.^{17,18} The recent EMPT-ANGINA trial is the first to report SGLT2i antianginal effects in patients with T2DM with coronary artery disease (CAD) and refractory angina, a population with

What Are the Clinical Implications?

Endothelial dysfunction is a hypothesized mechanism for coronary microvascular and vasomotor dysfunction (CMVD) in patients with angina and no obstructive coronary artery disease (ANOCA); however, mechanistically targeted treatment of endothelial dysfunction and translational studies are limited in this population. In this proof-of-concept, open-label clinical trial, patients with ANOCA with CMVD were treated with the SGLT2i (sodium-glucose co-transporter-2 inhibitor) empagliflozin for 90-days. SGLT2i was well tolerated in this population, with no serious AEs. We observed both a clinically and statistically significant improvement in angina, the predominant driver of morbidity in patients with ANOCA. Our novel in vitro serum model demonstrated that, compared with serum of no-CMVD controls, ANOCA-CMVD had a heightened incidence of endothelial dysfunction that was significantly reduced with SGLT2i treatment and rebounded following treatment discontinuation. Our findings suggest that SGLT2i may represent a novel therapeutic treatment in patients with ANOCA with CMVD that warrants investigation in future studies.

high prevalence of CMVD.¹⁹ Additionally, several rigorous in vivo and in vitro studies have demonstrated that SGLT2i reverse EnD.^{20–23} Specifically, SGLT2i reverse EnD through reduction in circulating inflammatory cytokines and leukocyte adhesion molecules, inhibition of excessive reactive oxygen species (ROS) production, and reduction in endothelial cell (EC) apoptosis with an increase in EC viability.^{20–23}

We conducted a proof-of-concept, open-label pilot trial to investigate the SGLT2i effects in ANOCA participants with CMVD confirmed by invasive coronary functional testing (ICFT). We assessed changes in angina, biomarkers of inflammation and oxidative stress, and indices of EnD leveraging a novel serum-induced in vitro model. We also compared these same measures in ANOCA-CMVD participants enrolled in the trial at baseline and no-CMVD controls from stored samples in our Coronary Microvascular and Vasomotor Dysfunction biorepository.²⁴ We hypothesized that SGLT2i would improve angina, biomarkers of inflammation, and indices of EnD in ANOCA participants with CMVD. Further, we hypothesized that ANOCA participants with CMVD would demonstrate increased measures of inflammation, oxidative stress, and indices of EnD compared with no-CMVD controls.

METHODS

The data, analytic methods, and study materials that support the findings of this study are available from the corresponding author on reasonable request to other researchers. Reporting

for this study is in line with the Strengthening the Reporting of Observational Studies in Epidemiology guidelines.

Study Overview

We conducted a proof-of-concept, open-label pilot trial in 11 ANOCA participants with CMVD treated with SGLT2i EMPA. The study was conducted between October of 2024 and April of 2025. The study was IRB-approved, and a Food and Drug Administration exemption was obtained for EMPA use in ANOCA participants with CMVD. ANOCA no-CMVD control blood samples were leveraged from an existing IRB-approved prospective CMVD biorepository of patients with ANOCA undergoing ICFT from a well-established CMVD program.²⁴

Study Population

All ANOCA participants enrolled in the pilot trial had undergone a clinically indicated ICFT for the diagnosis of CMVD. Inclusion criteria for the trial included (1) angina, (2) no obstructive CAD, and (3) CMVD diagnosed by ICFT using the bolus-thermodilution method. Angiographic inclusion criteria included (1) no obstructive CAD defined as coronary narrowing <50%, and fractional flow reserve ≥ 0.80 in all major epicardial arteries; (2) diagnosis of CMVD; and (3) on stable medical treatment with max tolerated therapy by CMVD endotype. CMD was defined by coronary flow reserve <2.5. Vasomotor dysfunction defined as any one of the following response to intracoronary acetylcholine: coronary EnD (CED) as constriction in proximal left anterior descending artery diameter of greater than or equal to 0% to mid dose acetylcholine (44 μ g acetylcholine), microvascular spasm as <90% constriction with angina and ischemic EKG changes in response to high dose acetylcholine (108 or 200 μ g acetylcholine), or coronary spasm as >90% constriction in response to any acetylcholine dose administered (44, 108, or 200 μ g) in the left circumflex or left anterior descending artery. ICFT thermodilution procedure performed was described in detail previously.²⁴

Exclusion criteria for the trial included (1) clinical diagnosis of T1DM or T2DM; (2) clinical diagnosis of heart failure; (3) history of coronary artery bypass graft; (4) severe valvular heart disease (any valve); (5) current treatment with SGLT2i; (6) impaired renal function (estimated glomerular filtration rate <60 mL/min/1.73 m²); (8) history of chronic or recurrent urinary tract infection or genital mycotic infections; (9) pregnancy or planned pregnancy in women of reproductive potential; (10) any clinical condition that might interfere with the trial protocol or the participant's ability to be compliant with the trial protocol; or (11) enrolled in another trial that might interfere with the current trial end points or procedures.

No-CMVD controls were leveraged from stored samples in the CMVD biorepository of consecutive patients undergoing clinically indicated ICFT and found to have no obstructive CAD and did not meet diagnostic criteria for CMVD. Patients were consented, and all samples were collected at the time of ICFT and processed using the same methods as the trial participants. Patients were referred to ICFT from a well-established clinical CMVD program as previously described in detail.²⁴ We excluded T1DM or T2DM, clinical diagnosis of heart failure, history of coronary artery bypass graft, and severe valvular disease (any valve) similar to the trial cohort from the no-CMVD control cohort.

Pilot Trial Data Collection

Trial measures (including clinical measurements, safety labs, and adverse events [AEs]) and patient-reported outcomes were only examined in ANOCA-CMVD participants. For all ANOCA-CMVD participants, data were collected at 3 in-person study visits (baseline, 45 days, and 90 days of EMPA treatment) and a phone visit for monitoring AEs. Additionally, in a subset of trial participants (N=5), a posttreatment discontinuation (PTD) in-person study visit was performed. Data collected included vitals, safety labs, urinary analysis, validated questionnaires, and blood collection. Vitals collected at each in-person visit included body mass index, systolic blood pressure, diastolic blood pressure, and heart rate. Safety labs collected and analyzed in real time included basic metabolic panel and urinary analysis. Participants were monitored for AEs from the time of consent to 30-days following discontinuation of study drug through phone visits.

Patient-reported outcomes were obtained using validated questionnaires including the Seattle Angina Questionnaire (SAQ) and the Duke Activity Status Index (DASI). SAQ summary score and SAQ subcomponent scores were calculated for physical limitation, angina stability, angina frequency, treatment satisfaction, and quality of life, with lower scores indicating worse status. DASI scores were calculated, with higher scores indicating higher functional capacity.²⁵ DASI scores were converted into metabolic equivalents (METs), and trial participants were divided into 2 groups using a METs <4.7 cutoff previously associated with adverse events.^{26,27}

Blood Collection and Processing

Blood was collected and processed at the time of study visits in the form of plasma, serum, and platelet-poor plasma. Serum was left at room temperature for 30 minutes to 2 hours post-collection to allow for clotting and was centrifuged at 400g (1500 rpm) for 10 minutes before storage. EDTA plasma was centrifuged at 400g (1500 rpm) for 10 minutes within 10 minutes of collection. Platelet-poor plasma was isolated using 2 successive centrifugations at 1900g for 10 minutes each. All samples were stored at -80°C. For no-CMVD control samples from the CMVD biorepository, blood was collected at the time of ICFT. Stored blood from the trial across various time points in ANOCA-CMVD and no-CMVD controls was used to examine circulating biomarker analyses, Flow Cytometry Analysis of Circulating Endothelial Microvesicles, and all in vitro sera-induced model measures (HCAEC Apoptosis and Viability, HCAEC eEV Release and VCAM-1 Expression, HCAEC Mitochondrial ROS) as detailed below.

Circulating Biomarker Analysis

Inflammatory cytokines were analyzed via multiplex ELISAs (RayBiotech, 40-plex, QAH-IMR-1-1) for each time point for ANOCA-CMVD trial participants (baseline, 45-days, 90-days, PTD) and at time of ICFT for no-CMVD controls. Biomarkers of interest included CRP (C-reactive protein), IL-6 (interleukin-6), TNF α (tumor necrosis factor- α), IL-1 β , endothelial selectin (E-selectin), thrombomodulin, ICAM-1 (intercellular adhesion molecule-1), VCAM-1 (vascular cell adhesion molecule-1), interleukin-10 (IL-10), and VEGF (vascular endothelial growth factor). Sandwich ELISAs for ROS and antioxidant

markers: total antioxidant capacity (AdipoGen Life Sciences, JAI-KPA-050), MPO (myeloperoxidase; RnD Systems, DMYE00B), MPO Activity Assay Kit (Abcam, ab105136), and F2 isoprostane (8-isoprostane, Cayman Chemicals, 516351). Assays were conducted and quantified in alignment with manufacturer guidelines.

Flow Cytometry Analysis of Circulating Endothelial Microvesicles

Endothelial microvesicles (eEVs) were analyzed as previously described and in accordance with the latest standardized extracellular vesicle (EV) enumeration publications by the ISTH-SCC-VB Working Group.²⁸ EVs were isolated using 200 μ L of platelet-poor plasma in HBSA without calcium and centrifuged at 20 000g for 20 minutes at 4°C at each time point, as previously described. Particle size distributions and enumeration were conducted using nanoparticle tracking analysis on a NanoSight NS300 instrument (Malvern, Worcestershire, United Kingdom), as previously described.²⁹ Flow cytometry detection of EC and endothelial progenitor cell eEVs was performed as described in the standardization of EVs utilizing a flow cytometry sub-micron particle size reference kit (Thermo Fisher, F13839) and a core-operated Beckman CytoFLEX S with 9 detectors and 4 lasers. EVs were quad-labeled for EC specific antibodies including Alexa Fluor 488-CD31 (Clone WM59, catalog no. 303110), PE-VCAM-1/CD106 (Clone STA, catalog no. 305806), Pac-Blue-CD63 (Clone H5C6, catalog no. 353012), and APC-annexin V (catalog no. 640920) for inflammatory eEVs, or CD31, PE-CD34 (Clone 581, catalog no. 343506), CD63, and annexin V (all from BioLegend) for anti-inflammatory eEVs and were quantified as previously described utilizing FlowJo software v10.10 (BD Life Sciences).

In Vitro Sera-Induced Model

For ANOCA-CMVD trial participants, sera at each time point were used (baseline, 45-days, 90-days, PTD), and for no-CMVD controls, sera were collected at the time of ICFT. All experiments were performed in cultured human coronary artery ECs (HCAECs; Lonza CC-2585), cultured in EGM-2 MV medium with SingleQuots supplements at 37°C with 5% CO₂. We used N=3 independent biological preparations, each run in triplicate for all assays, cultured in EGM-2 MV medium with SingleQuots supplements at 37°C with 5% CO₂. We used N=3 independent biological preparations, each run in triplicate for all assays.

HCAEC Apoptosis and Viability

HCAECs were treated with a standard curve of serum up to 20% (v/v) for 24 hours. Cells were seeded in 96-well plates (2.0×10⁵ cells/mL). Viability was measured using the CellTiter 96 MTS assay (Promega G3582); 100 μ L of media was incubated with 20 μ L MTS reagent for 1.5 hours, and absorbance at 490 nm was read on a Cytation5 plate reader. Percent viability was calculated relative to untreated controls. Apoptosis was quantified using the Click-iT TUNEL Alexa Fluor 488 assay (Thermo Fisher C10245). After serum treatment, cells were fixed, permeabilized, and labeled with the TdT reaction mixture, with DNase-treated wells as positive controls. DAPI counterstaining and imaging were performed on a Cytation5, and apoptosis was expressed as the percentage of TUNEL-positive nuclei across 16 high-power fields per well.

HCAEC eEV Release and VCAM-1 Expression

HCAECs were treated with serum for 24 hours. Supernatants were cleared by sequential centrifugation (1000g and 2500g), and eEVs were isolated at 20 000g for 20 minutes at 4°C. eEV size and number were assessed using a NanoSight NS300. VCAM-1 was quantified by ELISA (RnD DVC00), with supernatants diluted 1:20 and lysates 1:100, and normalized to total protein via Bradford assay.

HCAEC mROS

HCAECs were treated for 24 hours with 10% serum, with select wells pretreated for 1 hour with Mito-Tempol (5 μ mol/L). Cells were stained with MitoSOX Red, MitoTracker Green FM, and Hoechst 33 342 per manufacturer instructions (Invitrogen), incubated at 37°C for 30 minutes, washed, and imaged live on a Cytation5 (Agilent BioTek) using the appropriate imaging filter cubes. Mean fluorescence intensity per cell was quantified across 8 high-power fields per well using Agilent BioTek Gen5 software. Representative images were captured via an LSM710 confocal microscope (Carl Zeiss Inc) with a 10X objective.

Statistical Analysis

Group differences between ANOCA-CMVD at BL (N=11) and no-CMVD controls (N=6) were assessed using unpaired *t* tests for continuous variables, χ^2 tests for categorical variables, and 2-way ANOVAs where appropriate. ANOVAs for repeated measures where appropriate. Normally distributed data were presented as means±SD. Non-normally distributed data were summarized as medians (interquartile range) and compared using the Mann-Whitney *U* test.

In the ANOCA-CMVD trial, for participants treated with EMPA, to evaluate within-subject changes in biomarkers of inflammation, oxidative stress, and EnD, we applied restricted maximum likelihood linear mixed-effects models to account for repeated measures and the small sample size while including all time points (baseline, 45 days, 90 days, PTD). One participant withdrew from the study for reasons unrelated to the study at 38-days after completing visit 2. In line with the intent-to-treat principle, we included data from all participants that completed visits at baseline (N=11), 45 days (N=11), 90 days (N=10), and the subset that completed PTD (N=5). Time point (baseline, 45 days, 90 days, PTD) was modeled as a fixed effect with a random intercept for participants. When the main effect of time was significant, post hoc pairwise comparisons were performed using Tukey honestly significant difference. For 2-factor models, only significant interactions (eg, concentration × treatment) are reported with posttesting as appropriate. Missing data were not observed for the reported data in this study. Given the pilot, hypothesis-generating nature of our study, no formal power calculation was conducted prior to trial initiation. For clarity, figures denote statistically significant effects for posttesting, which was only conducted when main effects or significant interactions were found. Alpha was set at 0.05. Analyses were performed in GraphPad Prism (v10.1, GraphPad) and Stata (Stata19, StataCorp). The graphical abstract was created using FigureLabs text-to-figure function, with precise creation wording based on the results of the article and the coronary microvascular dysfunction literature. All software used is described in the [Major Resources Table](#) in the [Supplemental Material](#).



RESULTS

No CMVD Controls Compared With ANOCA-CMVD Clinical Trial Participants

There were no significant differences between no-CMVD controls (N=6) and ANOCA-CMVD clinical trial participants on baseline age, sex, and other demographics. Risk factors were also similar between the groups except for hyperlipidemia, which was more prevalent in the ANOCA-CMVD cohort. Patients with ANOCA-CMVD had a higher proportion of cardiac medications compared with no-CMVD controls (Table 1).

CMVD Clinical Trial Participant Characteristics

The majority of ANOCA-CMVD clinical trial participants were female (91%) with a mean age of 60.6 years (Table 1). In the subset (N=5) of trial participants who completed the additional in-person visit 4 (PTD), the mean time between visit 3 (90-day EMPA) and visit 4 was 47 ± 16.6 days. All trial participants had CED and additional concomitant CMVD endotypes, including CMD in 55% (N=6), microvascular spasm in 18% (N=2), and coronary spasm in 54% (N=6). The median time between ICFT CMVD diagnosis and trial enrollment was 20 months (interquartile range: 14, 31). On clinically performed transthoracic echo, the average ejection fraction (%) was 59.9 ± 4.5 , the average E/e' was 8.7 ± 2.6 , and the highest diastolic dysfunction grade observed was 1 in 22% (N=2) of participants.

Average clinical measures at each study visit are described in Table 2. One participant had a change in antianginal medication: the addition of a long-acting nitrate at day 72 of the study period. All other participants remained on a stable antianginal regimen throughout the trial period, confirmed at each study visit.

The most common AE related to the study drug was urinary tract infection symptoms (N=4) and urinary tract infection (N=2) confirmed by positive leukocytes in urinary analysis and/or treated with antibiotics. One participant had an estimated glomerular filtration rate of <60 mL/min/ 1.73^2 after 45 days of EMPA treatment that improved on repeat assessment and was within normal range at the 90-day visit. No serious AEs were observed or attributed to the study drug. An expanded description of AEs is outlined in Table S1.

CMVD Clinical Trial Participant Reported Outcomes

SAQ summary score and subcomponent scores in the ANOCA-CMVD clinical trial participants at each time point are shown in Table 2 and Figure 1. Changes in SAQ summary score and SAQ subcomponent scores in the ANOCA-CMVD trial participants from baseline to 45

Table 1. Demographics and Clinical Characteristics of No-CMVD Controls and ANOCA-CMVD Clinical Trial Participants at Baseline

Demographics	No CMVD (N=6)	CMVD BL (N=11)	P value
Age, y	49.5 $\pm$ 13.1	60.6 $\pm$ 13.3	0.11
Sex, n (%)			1.0
Female	6 (100)	10 (91)	
Male	0	1 (9)	
Race/ethnicity, n (%)			1.0
White	6 (100)	10 (91)	
Black	0	1 (9)	
Education, n (%)			0.88
High school or GED	1 (16)	3 (27)	
Some college	0	1 (9)	
Bachelors	1 (16)	2 (18)	
Masters degree or greater	2 (33)	2 (18)	
Declined to answer	2 (33)	3 (27)	
Income, n (%)			0.10
\$35 000–\$49 999	0	1 (9)	
\$50 000–\$99 999	0	5 (45)	
\$100 000 or more	3 (50)	1 (9)	
Declined to answer	3 (50)	4 (36)	
Comorbidities and CVD risk factors, n (%)			
Hypertension	2 (33)	7 (64)	0.23
Hyperlipidemia	4 (66)	11 (100)	0.04
Smoking status			
Former	3 (50)	2 (18)	0.16
Never	3 (50)	9 (82)	
Cardiac medications, n (%)			
Beta blockers	1 (16)	3 (27)	0.49
Calcium channel blockers	5 (83)	10 (91)	0.64
Long-acting nitrates	5 (83)	7 (63)	0.39
Short-acting nitrates	6 (100)	10 (91)	0.44
Ranolazine	0	5 (45)	0.04
Statins/PSK9i	3 (50)	10 (91)	0.05
RAASi	0	6 (54)	0.04
Aspirin	4 (66)	10 (91)	0.20

Means presented $\pm$ SD. ANOCA, angina with no obstructive coronary arteries; CMVD, coronary microvascular and vasomotor dysfunction; GED, general education degree; RAASi, renin angiotensin aldosterone system inhibitors.

and 90 days of EMPA treatment and from 90-days of EMPA treatment to PTD are detailed in Table S2. EMPA treatment improved angina as measured by an increase in SAQ summary score ($P=0.012$). Post hoc testing for the SAQ summary score revealed this difference was driven by an increase at 45 days ($P=0.004$) and at 90 days ($P=0.001$) of EMPA treatment, and there was no significant difference from 90 days of treatment to PTD ($P=0.99$). This effect remained consistent, excluding the 1 participant who had an anti-anginal regimen change in the study period (data not shown). 54% of the cohort

Table 2. Clinical Measurement and Patient-Reported Outcomes in ANOCA-CMVD Clinical Trial Participants Treated With Empagliflozin at Study Visits

	CMVD	CMVD	CMVD	CMVD	Main effect <i>P</i> value
Variable	BL (N=11)	45 EMPA (N=11)	90 EMPA (N=10)	PTD (N=5)	
Clinical measurements					
BMI, kg/m ²	26.6 (23.0–32.9)	26.9 (23.4–33.5)	26.7 (25.3–33.3)	26.7 (26.0–37.0)	0.42
Systolic blood pressure, mm Hg	130.1±17.4	131.0±11.2	137.5±13.6	118.2±14.8	0.11
Diastolic blood pressure, mm Hg	77.9±7.5	76.4±4.6	79.4±10.0	72.0±3.5	0.09
Lab measurements					
Sodium, mmol/L	140.2±2.4	139.6±1.7	139.9±2.2	NC	0.68
Potassium, mmol/L	4.2±0.4	4.3±0.4	4.3±0.4	NC	0.57
Glucose, mg/dL	100.5±19.7	92.4±8.4	88.3±7.1	NC	0.14
eGFR, mL/min/1.73m ²	84.6±15.3	79.2±18.3	85.8±16.9	NC	0.14
ALT, IU/L	28.9±21.3	25.6±16.0	32.3±31.6	NC	0.32
AST, IU/L	22.0±8.6	22.4±5.8	24.5±12.0	NC	0.34
BUN/creatinine ratio	16.1±3.4	16.5±2.3	16.9±3.0	NC	0.60
Patient-reported outcomes					
SAQ					
Summary score	58.2±12.0	68.0±12.3	72.2±15.0	71.6±17.8	0.01
Physical limitation score	63.0±19.5	66.3±20.3	76.2±18.3	72.2±23.6	0.16
Angina stability score	54.5±31.3	72.7±26.1	82.5±20.6	60.0±45.4	0.18
Angina frequency score	52.7±21.0	63.6±18.0	72.0±19.3	72.0±13.0	0.03
Treatment satisfaction score	91.7 (81.3–93.8)	93.8 (81.3–100)	90.6 (81.3–100)	94.0 (75.0–97.0)	0.50
Disease perception score	47.7±13.0	58.3±18.3	59.2±16.9	63.4±17.5	0.09
DASI	32.1±15.1	35.1±13.0	32.1±14.6	NC	0.53
DASI METS	6.6±1.7	6.9±1.5	6.6±1.6	NC	0.53
% of DASI METS <4.7	27%	0%	0%	NC	0.06

Means presented±SD. Medians described with IQR. ALT, alanine aminotransferase; AST, aspartate aminotransferase; BL, baseline; BMI, blood urea nitrogen; CMVD, coronary microvascular disease; eGFR, estimated glomerular filtration rate; EMPA, empagliflozin; METS, metabolic equivalent of task; NC, not collected; PTD, posttreatment discontinuation; and SAQ, Seattle Angina Questionnaire.

demonstrated a clinically significant improvement in SAQ summary score (>10-point increase) at 45-days and 70% of the cohort demonstrated clinically significant improvement at 90-days.

In SAQ subcomponents, there was an overall significant improvement in angina frequency ($P=0.03$); however, post-hoc comparisons for specific EMPA treatment time points were not significant. At baseline, 80% of the cohort experienced daily or weekly angina, which was reduced to 0% at 90 days of EMPA treatment. There were no significant differences in the other SAQ subcomponents, including physical limitation, angina severity, treatment satisfaction, or disease perception with treatment. Functional capacity, as measured by DASI and DASI METS for each study visit, is shown in Table 2. Analysis of DASI by METS associated with MACE showed that 27% of participants ($N=3$) had METS <4.7 at baseline, and this number was reduced to 0% at both 45 and 90 days of EMPA treatment. DASI score did not change between baseline and 45 or 90 days of EMPA treatment (Table S2).

Indices of EnD in No-CMVD Controls and ANOCA-CMVD Clinical Trial Participants Across Study Visits

Concentrations of circulating biomarkers and in vitro EnD biomarkers in no-CMVD controls and ANOCA-CMVD clinical trial participants at baseline are detailed in Table S3. Concentrations and changes in concentrations of circulating biomarkers and in vitro EnD indices in ANOCA-CMVD trial participants treated with EMPA at each study visit are detailed in Table 3 and Table S4.

Circulating Inflammation and Oxidative Stress Biomarkers and eEVs

Circulating markers of inflammation and oxidative stress are described in Figure 2, compared with no-CMVD controls, ANOCA-CMVD participants at baseline had higher concentrations of inflammatory markers (CRP, TNF α , IL-1 β), leukocyte adhesion markers (E-Selectin, thrombomodulin, ICAM-1, VCAM-1, VEGF), and ROS markers (MPO, MPO total activity, plasma 8-isoprostane),

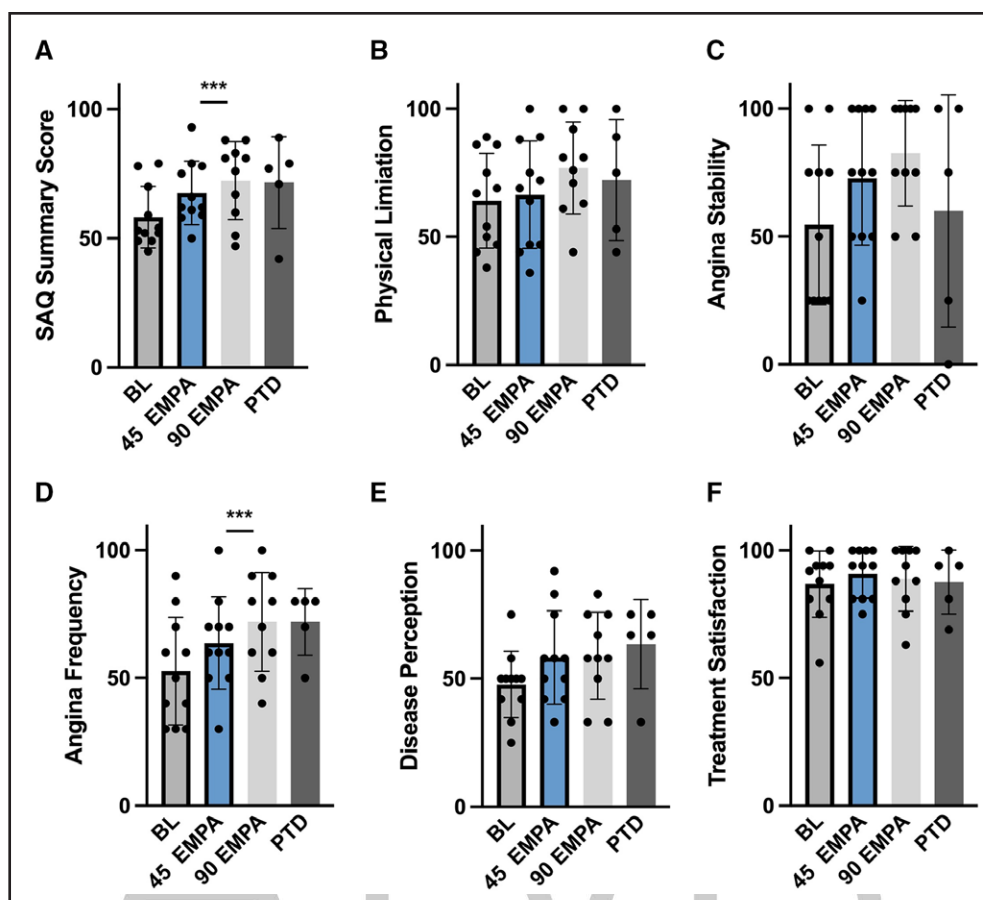


Figure 1. Angina changes in angina with no obstructive coronary arteries, coronary microvascular and vasomotor dysfunction (ANOCA-CMVD) clinical trial participants treated with empagliflozin (EMPA) at study visits.

A. Seattle Angina Questionnaire (SAQ) summary score; **(B)** SAQ physical limitation; **(C)** SAQ angina stability; **(D)** SAQ angina frequency; **(E)** SAQ disease perception; and **(F)** SAQ treatment satisfaction. Data from participants who completed each visit within the window are included for baseline (BL; N=11), 45 days (N=11), 90 days (N=10), and posttreatment discontinuation (PTD; N=5). Change over time was evaluated via restricted maximum likelihood. ***Denotes a significant difference between BL vs 45-days and BL vs 90-days using Tukey honestly significant difference post hoc testing. Data presented as means and SDs.

and lower concentrations of total antioxidant capacity and anti-inflammatory marker (IL-10; all $P < 0.002$). EMPA treatment reduced inflammatory, leukocyte adhesion, and oxidative stress markers (all $P < 0.0001$), and increased total antioxidant capacity ($P < 0.0001$) and IL-10 ($P < 0.0001$) at 45 and 90 days in the ANOCA-CMVD clinical trial participants. Between 90 days of EMPA treatment and PTD, inflammatory, leukocyte adhesion, and oxidative stress markers increased (all $P < 0.05$), and both total antioxidant capacity ($P = 0.004$) and IL-10 decreased ($P < 0.001$).

Circulating eEV concentrations are described in Figure 3. Compared with no-CMVD controls, CMVD participants at baseline had increased CD34⁺ endothelial repair-associated eEVs ($P < 0.001$), higher eEV median size ($P = 0.0002$), and CD31⁺/VCAM-1⁺ inflammation-associated eEVs ($P = 0.0002$). EMPA treatment significantly decreased the concentration of total circulating eEVs and eEV median size (all $P \leq 0.01$), increased CD34⁺ eEVs, and decreased CD31⁺/VCAM-1⁺ eEVs at 45 and

90 days (all $P < 0.001$) in the ANOCA-CMVD clinical trial participants. Between 90-day EMPA treatment and PTD, total eEVs, eEV median size, and CD31⁺/VCAM-1⁺ eEVs increased, while CD34⁺ eEVs decreased (all $P \leq 0.01$).

In Vitro Indices of EnD: EC Viability and Apoptosis

In vitro capacity to induce EC viability and apoptosis is detailed in Figure 4. There was a significant interaction between group (no-CMVD versus CMVD baseline) and EC viability ($P < 0.001$). Compared with no-CMVD control serum, HCAECs incubated with CMVD baseline serum induced lower EC viability at 5%, 10%, and 20% concentrations ($P < 0.01$ for all). There was a significant interaction between treatment time point and EC viability ($P < 0.001$) in the ANOCA-CMVD clinical trial participants. Compared with CMVD baseline serum, HCAECs incubated with EMPA-treated serum had higher EC

Table 3. Indices of Endothelial Dysfunction in ANOCA-CMVD Clinical Trial Participants Treated With Empagliflozin at Study Visits

Variable	CMVD BL (N=11)	CMVD 45 EMPA (N=11)	CMVD 90 EMPA (N=10)	CMVD PTD (N=5)	Main effect P value
Circulating biomarkers					
CRP, pg/nL	921.78±62.44	442.76±26.64	306.72±43.42	673.74±70.63	<0.0001
TNF-α, pg/nL	11.70±0.52	1.26±0.12	0.89±0.14	8.12±0.36	<0.0001
IL-1β, pg/nL	3.90±0.21	2.33±0.18	1.65±0.23	3.10±0.19	<0.0001
IL-6, pg/nL	9.88±3.81	2.78±0.17	2.16±0.15	8.16±0.43	<0.0001
E-Selectin, pg/nL	31 037.02±995.70	24 226.36±709.58	20 552.67±1 120.21	27 996.54±1 242.84	<0.0001
Thrombomodulin, pg/nL	11 244.20±967.06	7602.26±280.86	6812.66±301.81	9446.86±532.49	<0.0001
ICAM-1, pg/nL	90 040.42±1317.72	75 675.61±1309.82	61 136.56±3751.73	86 179.14±3530.59	<0.0001
VCAM-1, pg/nL	374.95±20.05	74.10±11.04	57.58±3.45	282.94±5.55	<0.0001
VEGF, pg/nL	6.19±0.33	4.11±0.19	3.66±0.26	5.34±0.32	<0.0001
MPO, pmol/L	119.41±3.57	49.25±3.38	33.91±2.31	83.50±5.82	<0.0001
MPO total activity, mU/mg	30.29±0.52	16.78±0.43	14.44±0.50	24.58±0.66	<0.0001
8-Isoprostane, pg/nL	326.75±3.76	233.45±6.69	212.74±5.06	293.48±6.20	<0.0001
PAO, uM/L reducing power	546.43±17.89	708.17±7.11	835.43±14.13	623.61±19.69	<0.0001
IL-10, pg/nL	0.19±0.02	9.41±0.81	9.69±1.16	0.66±0.09	<0.0001
Circulating eEVs					
eEV size, nm	84.28±4.55	63.40±9.98	53.00±3.01	76.91±4.02	<0.0001
eEV concentration, particles/mL	3.04×10 ⁸ ±1.79×10 ⁹	1.02×10 ⁸ ±1.88×10 ⁸	2.96×10 ⁷ ±4.04×10 ⁷	9.89×10 ⁸ ±5.09×10 ⁸	0.007
VCAM ⁺ eEVs, %	37.32±1.48	16.53±1.01	8.66±0.71	26.76±0.81	<0.0001
CD34 ⁺ eEVs, %	7.52±0.29	22.91±1.06	26.76±0.88	12.18±0.93	<0.0001
Sera induced HCAEC in vitro model					
eEV Size, nm	105.26±5.88	72.28±7.58	67.20±8.50	88.20±2.48	0.001
eEV concentration, particles/mL	5.82×10 ⁸ ±3.10×10 ⁸	4.71×10 ⁷ ±9.66×10 ⁶	2.82×10 ⁷ ±6.58×10 ⁶	3.42×10 ⁸ ±4.50×10 ⁷	<0.0001
VCAM-1 supernatant concentration, pg/mL	66.10±8.74	43.60±4.50	35.87±2.62	54.04±4.29	<0.0001
VCAM-1 lysate concentration, pg/g protein	588.99±60.23	337.74±22.90	328.96±25.89	450.60±13.58	<0.0001
Viable cells (%) 5% sera	85.78±2.27	92.32±1.40	93.11±1.23	87.13±0.67	<0.0001*
Viable cells (%) 10% sera	54.50±0.60	72.09±0.75	79.13±1.72	63.59±2.10	
Viable cells (%) 20% sera	37.39±4.17	57.78±0.65	64.12±2.54	47.85±2.03	
Apoptotic cells (%) 5% sera	10.66±0.69	8.33±2.00	6.14±0.51	9.54±0.49	<0.0001*
Apoptotic cells (%) 10% sera	25.47±1.33	15.58±1.73	13.89±1.03	21.54±1.04	
Apoptotic cells (%) 20% Sera	47.73±2.29	29.57±2.45	22.93±1.15	36.00±1.93	
MitoSOX fluorescence, a.u	11.73×10 ⁵ ±0.30×10 ⁵	2.10×10 ⁵ ±0.19×10 ⁵	2.05×10 ⁵ ±0.11×10 ⁵	7.79×10 ⁵ ±0.38×10 ⁵	<0.0001

Means presented±SD. ANOCA, angina with no obstructive coronary arteries; a.u, arbitrary units; CD34, cluster of differentiation 34; CMVD, coronary microvascular and vasomotor dysfunction; CRP, C-reactive protein; E-Selectin, endothelial selectin; EV, extracellular vesicle; HCAEC, human coronary artery endothelial cell; ICAM-1, intercellular adhesion molecule 1; IL, interleukin; MPO, myeloperoxidase; PAO, plasma antioxidant capacity; TM, thrombomodulin; TNF, tumor necrosis factor-α; VCAM-1, vascular cell adhesion molecule 1; VEGF, vascular endothelial growth factor.

*P value denotes the interaction effect for concentration and time.

viability at 45 days and 90 days at 5%, 10%, and 20% concentrations ($P<0.001$ for all). Compared with 90-day EMPA-treated sera, HCAECs incubated with PTD sera had lower EC viability at 5%, 10%, and 20% concentrations ($P<0.001$ for all).

There was a significant interaction between group (no-CMVD versus CMVD baseline) and EC apoptosis ($P<0.001$). Compared with no-CMVD control serum, HCAECs incubated with CMVD-BL serum had lower EC

apoptosis at 5%, 10%, and 20% ($P<0.001$ for all). Compared with CMVD –baseline serum, HCAECs incubated with EMPA-treated serum had lower EC apoptosis at 45 days and 90 days at 5%, 10%, and 20% concentrations ($P<0.001$ for all except $P=0.02$ at 5% at 45 days) in the ANOCA-CMVD clinical trial participants. Compared with 90 days EMPA treated sera, HCAECs incubated with PTD sera had higher EC apoptosis (5% $P=0.02$, 10% $P=0.001$, 20% $P<0.001$).

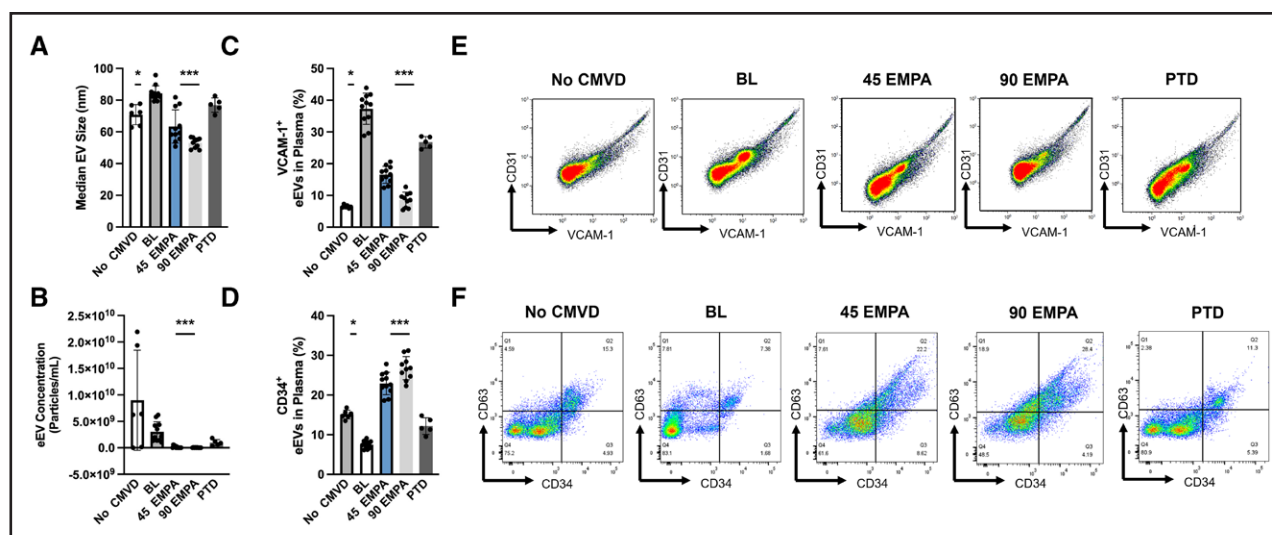


Figure 2. Concentrations of circulating inflammatory and oxidative stress biomarkers in no coronary microvascular and vasomotor dysfunction (CMVD) controls and angina with no obstructive coronary arteries CMVD (ANOCA-CMVD) clinical trial participants treated with Empagliflozin at study visits.

A, CRP (C-reactive protein); **(B)** TNF α (tumor necrosis factor alpha); **(C)** IL (interleukin)-1 β ; **(D)** IL-6; **(E)** E-Selectin; **(F)** thrombomodulin; **(G)** ICAM-1 (intercellular adhesion molecule 1); **(H)** VCAM-1 (vascular cell adhesion molecule 1); **(I)** VEGF (vascular endothelial growth factor); **(J)** plasma myeloperoxidase; **(K)** plasma myeloperoxidase total activity; **(L)** plasma 8-isoprostane; **(M)** plasma total antioxidant capacity (PAO); and **(N)** IL-10. Data from no-CMVD controls (N=6) and participants that completed each visit within the window baseline (BL; N=11), 45 days (N=11), 90 days (N=10), and posttreatment discontinuation (PTD; N=5) are included for each measurement. *Denotes a significant difference between No-CMVD vs CMVD-BL compared via unpaired *t* test. Change over time was evaluated via restricted maximum likelihood (REML) for ANOCA CMVD participants. ***Denotes a significant difference between BL vs 45-days, BL vs 90-days, and 90 days vs PTD among CMVD clinical trial participants via Tukey honestly significant difference. All data are presented as means and SDs. eEV indicates endothelial microvesicles.

In Vitro Indices of ED: eEV Release and mROS

In vitro effects on eEV release and VCAM are summarized in Figure 4. Compared with no-CMVD control serum, HCAECs incubated with CMVD baseline serum induced release of larger EVs based on median size and higher concentrations of total EVs and VCAM in the supernatant and lysate ($P<0.001$ for all). Compared with CMVD baseline serum, HCAECs incubated with EMPA-treated serum had smaller EVs based on median size and lower total EVs at 45 and 90 days ($P<0.01$), which increased at PTD compared with 90 days ($P=0.016$ and $P=0.02$, respectively) in the ANOCA-CMVD clinical trial participants. Compared with CMVD baseline serum, HCAECs incubated with EMPA-treated serum had lower VCAM concentrations in supernatant and lysate at 45 days and 90 days ($P<0.001$ for all), which increased PTD ($P=0.07$ and $P=0.003$, respectively).

In vitro effects on mitochondrial ROS are summarized in Figure 5. Compared with no-CVMD controls, HCAECs incubated with CMVD baseline serum had MitoSOX staining fluorescent intensity ($P<0.001$) consistent with mitochondrial reactive oxygen species (mROS) activity. Compared with CMVD baseline serum, HCAECs incubated with EMPA-treated serum had lower MitoSOX staining fluorescent intensity at 45 days and at 90 days ($P<0.001$ for both); and MitoSOX staining fluorescent intensity increased from 90 days to PTD ($P=0.001$) in

the ANOCA-CMVD clinical trial participants. In our control experiment, mROS activity was reduced in ANOCA-CMVD at baseline when the mitochondrial-targeted antioxidant mito-Tempol (5 $\mu\text{mol/L}$) was added to culture ($P<0.001$), suggesting specificity of staining.

DISCUSSION

In this proof-of-concept open-label pilot trial of ANOCA participants with ICFT-diagnosed CMVD treated with EMPA, we demonstrate SGLT2i treatment results in clinically significant angina improvement. We also demonstrate heightened circulating and serum-mediated drivers of EnD in patients with ANOCA with CMVD, as compared with no-CMVD controls, that improve with SGLT2i. Specifically, SGLT2i (1) reduced circulating markers of inflammation and oxidative stress, while increasing anti-inflammatory and antioxidant capacity; (2) normalized eEV phenotype by lowering circulating CD31⁺/VCAM-1⁺ inflammatory eEVs and increasing CD34⁺ progenitor eEVs; and (3) reversed indices of EnD in a novel in vitro serum-induced model; all of which rebound with treatment discontinuation.

To our knowledge, this is the first study to demonstrate improvement in angina with SGLT2i in an ANOCA population with invasively diagnosed CMVD. Our observed anti-anginal effects are consistent with a recent large

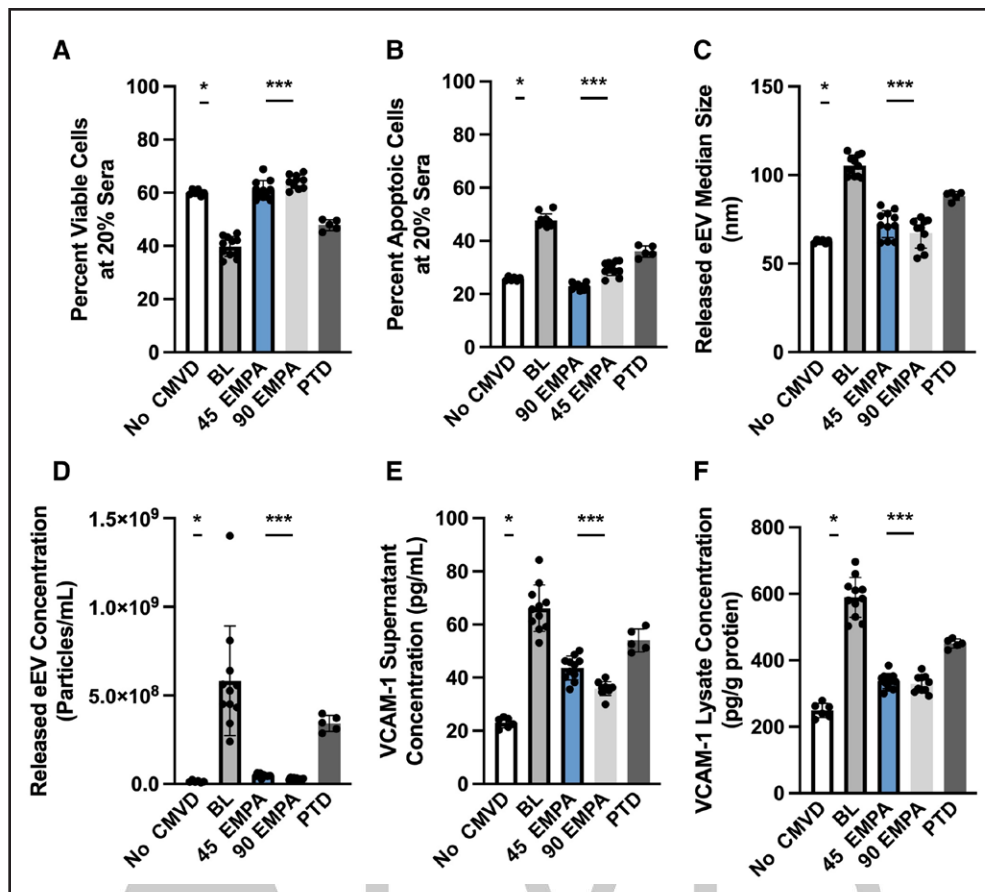


Figure 3. Circulating endothelial microvesicles (eEVs) in no coronary microvascular and vasomotor dysfunction (CMVD) controls and angina with no obstructive coronary arteries CMVD (ANOCA)-CMVD clinical trial participants treated with EMPA (empagliflozin) at study visits.

A, Median microvesicle (extracellular vesicle [EV]) size; **B**, concentration of circulating CD31⁺ and VCAM-1⁺ double-positive EVs in serum (%); **C**, representative flow cytometry forward scatter plots for CD31⁺ and VCAM-1⁺ EVs across treatment time points; **D**, concentration of circulating EV in serum; **E**, concentration of circulating CD63⁺ and CD34⁺ double-positive EVs in serum (%). **F**, Representative flow cytometry forward scatter plots for CD63⁺ and CD34⁺ EVs across treatment time points. Data from no-CMVD controls (N=6) and participants that completed each visit within the window baseline (BL; N=11), 450 days (N=11), 90 days (N=10), and posttreatment discontinuation (PTD; N=5) are included for each measurement. *Denotes a significant difference between No-CMVD vs CMVD-BL compared via unpaired *t* test. Change over time was evaluated via restricted maximum likelihood (REML) for ANOCA CMVD participants. ***Denotes a significant difference between BL vs 45 days, BL vs 90 days, and 90 days vs PTD among CMVD clinical trial participants via Tukey honestly significant difference. All data are presented as means and SDs.

placebo-controlled trial that demonstrated EMPA treatment significantly reduced anginal symptoms in patients with T2DM with obstructive CAD and refractory angina.¹⁹ In patients with ANOCA, angina has a significant impact on morbidity as it leads to recurrent emergency room visits and hospitalizations; therefore, the potential of EMPA to reduce the angina burden has a significant impact that warrants further investigation.³⁰ Although not consistent across all studies, SGLT2i treatment has demonstrated improvement in functional capacity as measured by VO_2 and 6-minute walk time; however, we did not demonstrate a significant improvement in patient-reported functional capacity as measured by DASI in this study.^{31,32} We hypothesize this is largely because the functional capacity in our study population was high at baseline, limiting treatment effects. In our study, only 30% of clinical trial

participants had DASI-estimated METs below 4.7 at baseline, which is the threshold previously shown to be associated with adverse events in patients with ANOCA; and this was reduced to 0% after EMPA treatment at both 45- and 90-day time points.²⁶ Therefore, future studies on the effects of SGLT2i treatment on objective measures of functional capacity in a larger cohort of ANOCA patients with CMVD are needed to determine efficacy.

Our data suggest 3 potential mechanistic theories that may explain the observed benefit of EMPA on indices of EnD in ANOCA participants with CMVD. First, anti-inflammatory modulation: EMPA rapidly suppresses systemic inflammation and leukocyte adhesion signaling by reducing circulating inflammatory cytokines and leukocyte adhesion molecules (IL-6, IL-1 β , TNF α , ICAM-1, VCAM-1,

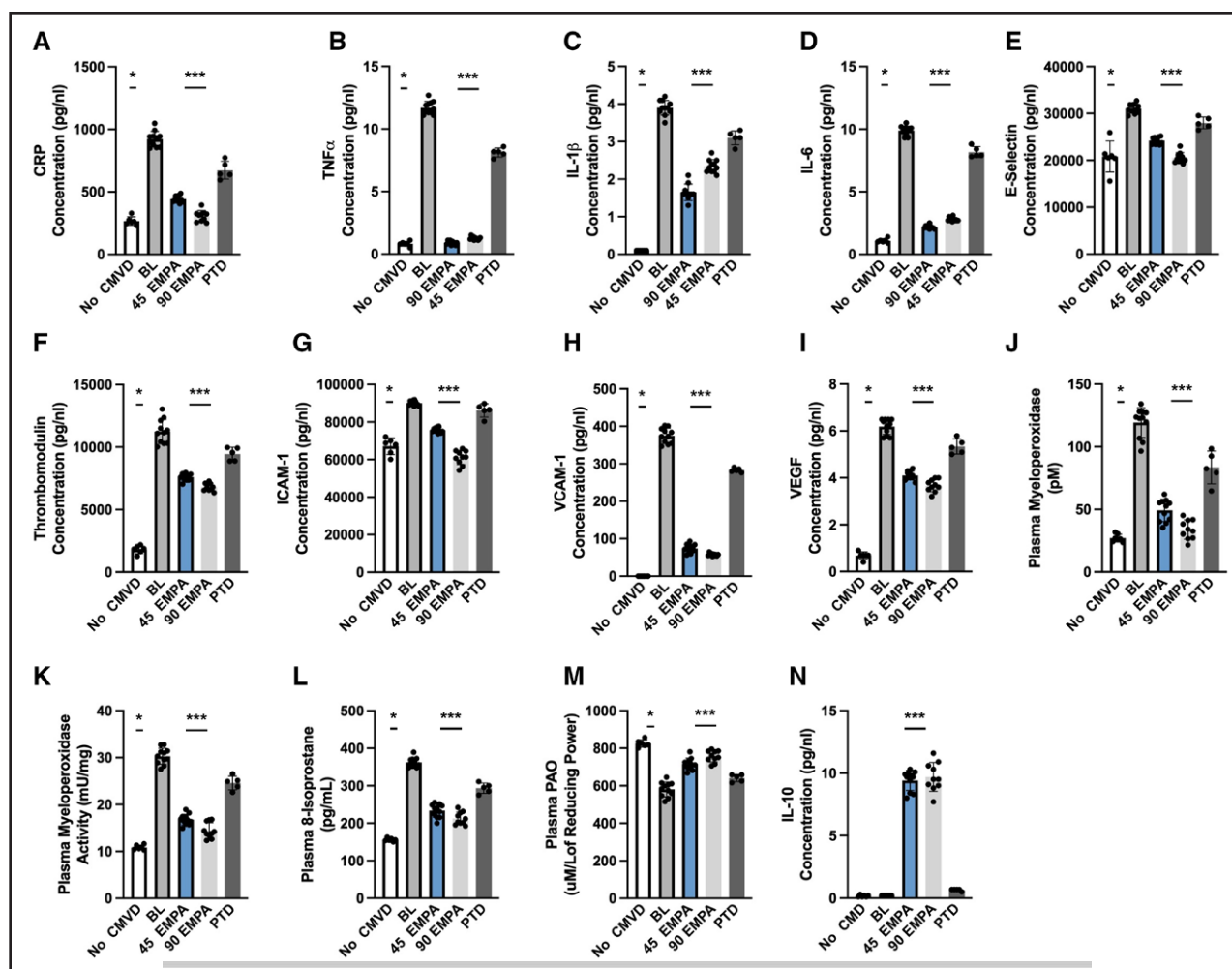


Figure 4. In vitro human coronary artery endothelial cells (HCAEC) serum-induced model of endothelial dysfunction markers in no coronary microvascular and vasomotor dysfunction (CVMD) controls and angina with no obstructive coronary arteries (ANOCA)-CMVD clinical trial participants treated with EMPA (empagliflozin) at study visits.

A, Percent of apoptotic cells at 20% sera; **B**, percent of viable cells at 20% sera. **C**, Median microvesicle (extracellular vesicle [EV]) size released by HCAECs; **D**, concentration of EV released by HCAECs (%); **E**, VCAM (vascular cell adhesion molecule)-1 in supernatant released by HCAECs; **F**, VCAM-1 released in cell lysate induced by incubation of HCAEC with 10% serum. Data from no-CMVD controls (N=6) and participants that completed each visit within the window baseline (BL; N=11), 45 days (N=11), 90 days (N=10), and posttreatment discontinuation (PTD; N=5) are included for each measurement. *Denotes a significant difference between No-CMVD vs CMVD-BL compared via unpaired *t* test. Change over time was evaluated via restricted maximum likelihood (REML) for ANOCA CMVD participants. ***Denotes a significant difference between BL vs 45-days, BL vs 90 days, and 90 days vs PTD among CMVD clinical trial participants via Tukey honestly significant difference. All data are presented as means and standard deviations.

E-selectin, CRP) and increasing anti-inflammatory signaling (IL-10), directly attenuating a major upstream driver of CMVD.^{33–35} Second, reversal of EnD through modulation of eEVs: EMPA decreased inflammatory CD31⁺/VCAM-1⁺ eEVs while increasing CD34⁺ repair-associated eEVs, a reciprocal shift suggesting suppression of endothelial activation together with enhancement of endogenous repair signaling in ECs. Moreover, reduced EC apoptosis combined with increased progenitor-type eEVs creates a biologic milieu that favors microvascular integrity and is congruent with prior reports of SGLT2i mobilizing endothelial progenitor signals and erythropoietic/reparative factors resulting in reversal of EnD.^{36–38} We also observed that EMPA reduced the size and concentration of eEVs

released in culture in our in vitro model; a change linked to lower cardiovascular risk in prior studies.^{39,40} Third, reversal of EnD through mitochondria-centered effects: serum from CMVD participants induced mROS and EC apoptosis in HACECs, which were attenuated by EMPA-treated serum. Prior work has demonstrated the potential of SGLT2i to restore nitric oxide signaling and improve cardiomyocyte relaxation, processes indirectly or directly related to mROS signaling.^{21,41,42} The transient nature of the EMPA effects, as demonstrated by rebound PTD, suggests that sustained therapy may be required for durable reversal of EnD.

It is important to note that these mechanisms are not mutually exclusive and may act synergistically with

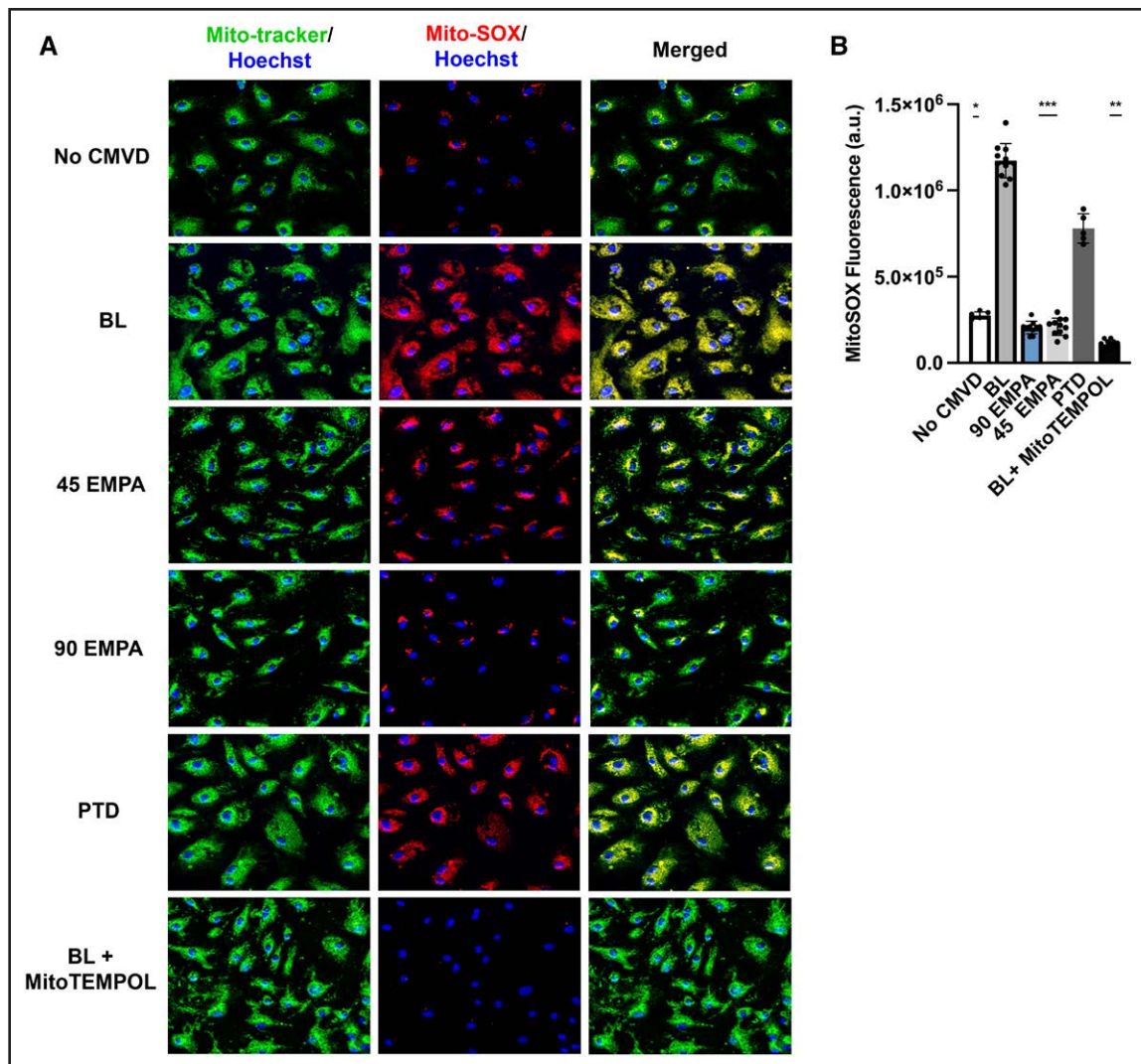


Figure 5. In vitro human coronary artery endothelial cells (HCAEC) serum-induced model MitoSOX release in no coronary microvascular and vasomotor dysfunction (CMVD) controls and angina with no obstructive coronary arteries (ANOCA) CMVD clinical trial participants treated with EMPA (empagliflozin) at study visits.

A, MitoSOX red fluorescence staining colabeled with for nuclei. Representative images across groups taken at 10x; blue staining indicates Hoechst (405 nm excitation), Mito-Tracker Green (490nm excitation), and red indicates MitoSOX staining (594 nm excitation). **B**, MitoSOX Red Fluorescence in arbitrary units (a.u.) $\times 10^5$ in HCAECs following incubation with serum. Data from no-CMVD controls (N=6) and participants that completed each visit within the window baseline (BL; N=11), 45 days (N=11), 90 days (N=10), and posttreatment discontinuation (PTD; N=5) are included for each measurement. *Denotes a significant difference between No-CMVD vs CMVD-BL compared via unpaired *t* test. Change over time was evaluated via restricted maximum likelihood (REML) for ANOCA CMVD participants. ***Denotes a significant difference between BL vs 45 days, BL vs 90-days, and 90 days vs PTD among CMVD clinical trial participants via Tukey honestly significant difference. All data are presented as means and SDs.

metabolic and hemodynamic effects of SGLT2i. To date, ECs are thought to express minimal functional SGLT2 receptors; therefore, some of the effects may involve off-target interactions (eg, SGLT1) or be mediated indirectly via hemodynamic effects and metabolic shifts.^{43,44} SGLT2i hemodynamic effects such as reduction in arterial stiffness, arterial blood pressure, afterload, and filling pressures (preload) through natriuresis could affect the observed improvements in angina and indices of EnD; however, there were no significant changes in blood pressure or weight in our small trial.^{45,46} SGLT2i-induced

metabolic shifts such as an increase in free fatty acids and ketogenesis-mediated beta-hydroxybutyrate can modulate inflammation, mitochondrial function, and substrate utilization in the heart, which warrants investigation in the follow-up trial to this pilot study.^{47–49}

To date, mechanistic studies in ANOCA patients with CMVD are limited. We demonstrated evidence of proinflammatory eEVs (VCAM-1⁺) and progenitor repair eEVs (CD34⁺) in ANOCA participants with CMVD. This is of importance since several in vitro studies have shown that stimulation of eEV release by inflammatory cytokines

(TNF α , IL-1 β , IL-1 α) can induce reduced capacity for vasodilation (eNOS uncoupling) and increased oxidative stress promoting EnD in healthy ECs.^{50–52} Additionally, circulating eEVs have been associated with increased MACE and mortality risk and used as clinical measures of EnD.^{53–55} Future studies isolating eEVs pre- and post-treatment are needed to explore the role of eEVs in SGLT2i effects.

In this ANOCA trial, all participants had CED and concomitant CMD and/or spasm diagnosed by ICFT. EnD is a key pathophysiologic pathway across all CMVD endotypes.^{11,56} We demonstrate this using a biologically valid serum-induced in vitro model. Compared with no-CMVD control serum, serum of ANOCA-CMVD trial participants at baseline induced EnD as reflected by higher HCAEC-induced release of eEVs, VCAM-1⁺, EC apoptosis, and mROS. Additionally, consistent with prior work, patients with ANOCA with CMVD had higher concentrations of circulating biomarkers of inflammation and oxidative stress. This included IL-6 and TNF α , which in the iPOWER study were associated with echocardiography-measured coronary flow velocity reserve.⁵⁷ Similarly, a small study showed upregulation of whole-blood transcriptional (RNA-sequencing) signatures associated with cytokine storm, ROS production, and inflammasome signaling in 7 ANOCA participants with CMVD compared with 21 participants without CMVD, consistent with our results.⁵⁸ Understanding the pathophysiological mechanism of CMVD in patients with ANOCA is critical to develop and subsequently investigate targeted treatments.

We hypothesize that the improvements in angina observed in this cohort of ANOCA-CMVD participants are due to improvement in CMVD. Prior work from WISE demonstrated that angina is related to underlying CMVD in patients with ANOCA. Our in vitro model demonstrated reversal of indices of EnD, which leads us to hypothesize that reversal of CED present in all participants may be a key driver of the improvement in angina. In addition to CED, all participants had concomitant CMD and spasm, so it is possible that SGLT2i also had an effect on these CMVD endotypes. Our hypothesis is further supported by animal studies demonstrating that SGLT2i treatment improves measures of CED and CMD. Also, a recent study in patients with T2DM showed a significant improvement in myocardial blood flow (noninvasive measure of CMD) at both 4-weeks and 4-years of Dapagliflozin treatment.⁵⁹ Similarly, the recently completed SMILE trial of 33 ANOCA participants, 10 of whom had CMD, found that 12-week treatment with dapagliflozin improved CMD as measured by stress-MRI-derived myocardial perfusion reserve index (published as an abstract).⁶⁰ Future larger randomized, placebo-controlled trials are needed to determine if the observed improvement in angina in ANOCA participants is due to change in CMVD with pre- and post-treatment ICFT measurements. The impact of future SGLT2i studies on

CMVD extends beyond ANOCA, as CMVD affects up to 79% of patients with heart failure with preserved ejection fraction and 40% of those with refractory angina postintervention.^{61–64}

Strengths and Limitations

This study has strengths and limitations. Strengths of the study include the use of gold-standard invasive diagnosis of CMVD using ICFT in patients with ANOCA, a SAQ-validated questionnaire and a validated in vitro HCAEC serum model. Additional strengths are the exclusion of participants with comorbid conditions (T2DM and heart failure) to remove confounding effects linked to glucose control and volume from the effects of SGLT2i. The longitudinal design allowed for evaluation of time-dependent effects of SGLT2i treatment and PTD on angina, inflammation, oxidative stress, and indices of EnD.

Despite the strengths, our study is not without limitations. First, it is important to note that we did not measure direct improvements in CMVD measures, and all indices of EnD were measured in vitro. Our findings suggest that discontinuation of EMPA results in regression back toward baseline in markers of inflammation, oxidative stress, and indices of EnD. However, the study was not long enough to observe whether the markers of EnD regress completely back to baseline nor whether there is a time-dependent loss of anti-anginal benefit. The duration and small size of the study limit the ability to determine the effects of treatment on clinical outcomes. Additionally, due to the presence of mixed CMVD endotypes and small sample size, we are unable to dissect treatment response by CMVD endotype. Notably, our cohort is predominantly female, limiting extending our findings can be extended to male patients. Notwithstanding these limitations, this small, open-label and hypothesis-generating trial did demonstrate significant improvement in angina, markers of inflammation and oxidative stress, and indices of EnD in vitro.

CONCLUSIONS

In this proof-of-concept, open-label clinical trial of patients with ANOCA with CMVD treated with SGLT2i EMPA, we observed a significant improvement in angina. ANOCA-CMVD participants have heightened inflammation, oxidative stress, and indices of EnD as compared with no-CMVD controls, that was significantly reduced with EMPA treatment and rebounded following treatment discontinuation. Our study indicates that SGLT2i is a potential novel therapy for patients with ANOCA-CMVD that warrants further investigation.

ARTICLE INFORMATION

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Disclosures

C.N. Bairey Merz serves on the board of directors and holds stock in iRhythm. The other authors report no conflicts.

Supplemental Material

Tables S1–S4
Major Resource Table

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